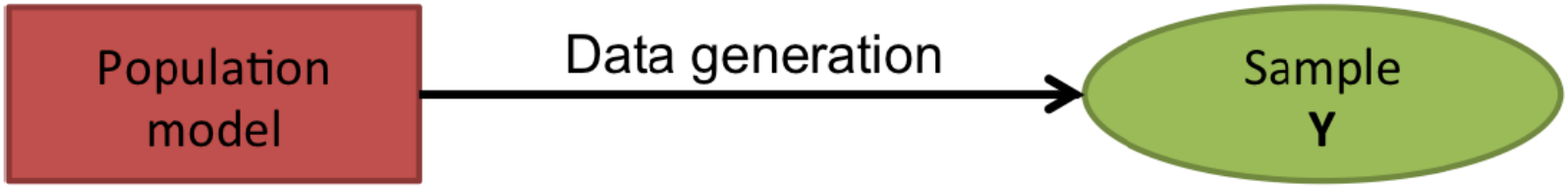
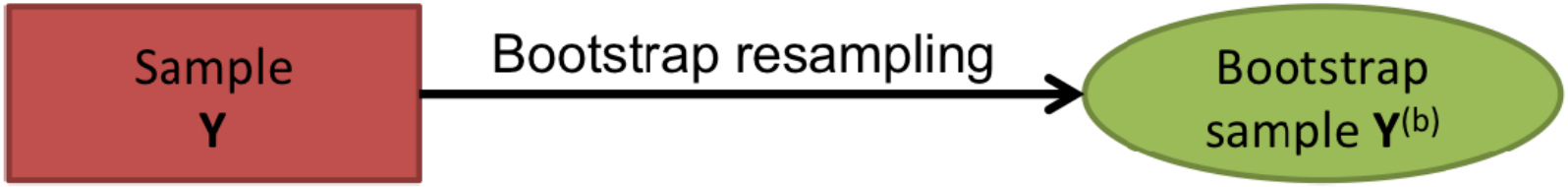
Bootstrap vs Bayesian Bootstrap

An analogy

The following analogy is helpful to keep in mind. For the sampling distribution we have:



Bootstrapping treats the sample $\mathbf{Y}$ as if it represents the true population model:



Bootstrap

- Denote the original data $y = (y_1, \ldots, y_n)$
- Denote a bootstrap sample $y^b = (y_1^b, \ldots, y_n^b)$
- The mean of the bootstrap sample is given by

$$m_b = \frac{1}{n} \sum_{i=1}^{n} y_i^b.$$

- It is the distribution m_b of over a number of bootstrap replications that is used to approximate the sampling distribution from the unknown population.

Bootstrap

- In each bootstrap sample, each observation occurs anywhere from 0 to n times.
- Let h_i^b denote the number of times y_i occurs in the bootstrap sample,

$$h_i^b \in \{0, 1, \ldots, n-1, n\} \text{ and } \sum_{i=1}^{n} h_i^b = n$$

- We can re-express the mean of the bootstrap sample as

$$m_b = \sum_{i=1}^{n} w_i^b \, y_i \, .$$

where $w_i^b = h_i^b / n$.

Bootstrap

The most standard form of the bootstrap, due to Efron (1979), involves selecting *bootstrap samples* $\mathbf{x}^* = (x_1^*, x_2^*, \ldots, x_n^*)$ by randomly sampling n times with replacement from the original sample $x_1, x_2, \ldots, x_n$. The idea is to select a large number B of these bootstrap samples $\mathbf{x}^{*1}, \mathbf{x}^{*2}, \ldots, \mathbf{x}^{*B}$. If $\hat{\theta}(\mathbf{x})$ is an estimate based on the original sample $\mathbf{x}$, then $\hat{\theta}(\mathbf{x}^{*b})$ is the corresponding estimate based on the bth bootstrap sample. The estimate $\hat{\theta}(\mathbf{x})$ could be, for instance, the mean or the median. By studying how the $\hat{\theta}(\mathbf{x}^{*1}), \hat{\theta}(\mathbf{x}^{*2}), \ldots, \hat{\theta}(\mathbf{x}^{*B})$ are distributed, we learn about the distribution of $\hat{\theta}(\mathbf{x})$.

Standard Error

In particular, Efron (1982, p. 28) suggests that the standard error of $\hat{\theta}(\mathbf{x})$ be estimated by

$$\widehat{\mathrm{SE}}[\hat{\theta}(\mathbf{x})] = \left\{ \frac{1}{B-1} \sum_{b=1}^{B} [\hat{\theta}^{*b}(\mathbf{x}) - \hat{\theta}^{**}(\mathbf{x})]^2 \right\}^{\frac{1}{2}}$$

where $\hat{\theta}^{**}(\mathbf{x}) = \sum_{b=1}^{B} \hat{\theta}^{*b}(\mathbf{x})/B$.

Two approaches to building 95% confidence intervals

- *The normal interval*: $[T(\mathbf{Y}) - 1.96\widehat{\mathsf{SE}}_{\mathsf{BS}}, T(\mathbf{Y}) + 1.96\widehat{\mathsf{SE}}_{\mathsf{BS}}]$. This approach assumes that the sampling distribution of $T(\mathbf{Y})$ is normal, and uses the standard error accordingly.
- *The percentile interval*: Let T_q be the q'th quantile of the bootstrap distribution. Then the 95th percentile bootstrap interval is: $[T_{0.025}, T_{0.975}]$.

Failure of the Bootstrap

- Bootstrap is not immune to yielding inaccurate results; for example, for extreme value statistics (e.g., the maximum or minimum of the data), the bootstrap can yield badly biased estimates.
- Suppose that $X_1, \ldots, X_n$ from Unif[0, θ] and we try to bootstrap the MLE to construct a confidence interval for θ.
- The MLE is the largest observation $X_{(n)}$. This point is contained in the bootstrap sample with probability $1 - (1 - 1/n)^n \approx .63.$

Failure of the Bootstrap

- The bootstrap distribution puts mass 0.63 at the single point $X_{(n)}$.
- But we know that $n(X_{(n)} - \theta)$ follows an exponential distribution.
- So the bootstrap distribution does not resemble the true distribution.
- “m out of n” bootstrap may work.

Example

- Suppose $X_1, \ldots, X_n$ are i.i.d. from Unif[0, θ]

$$T_n = n(\theta - X_{(n)})$$

$$T_n^* = n(X_{(n)} - X_{(n)}^*)$$

- However, the conditional distribution of T_n^* given $X_{(n)}$ does not converge to the Exp(θ).

Proof

$$
\begin{aligned}
P_{F_n}(T_n^* \leq t) &\geq P_{F_n}(T_n^* = 0) \\
&= P_{F_n}(X_{(n)}^* = X_{(n)}) \\
&= 1 - P_{F_n}(X_{(n)}^* < X_{(n)}) \\
&= 1 - \left(\frac{n-1}{n}\right)^n \\
&\overset{n\to\infty}{\longrightarrow} 1 - e^{-1}
\end{aligned}
$$

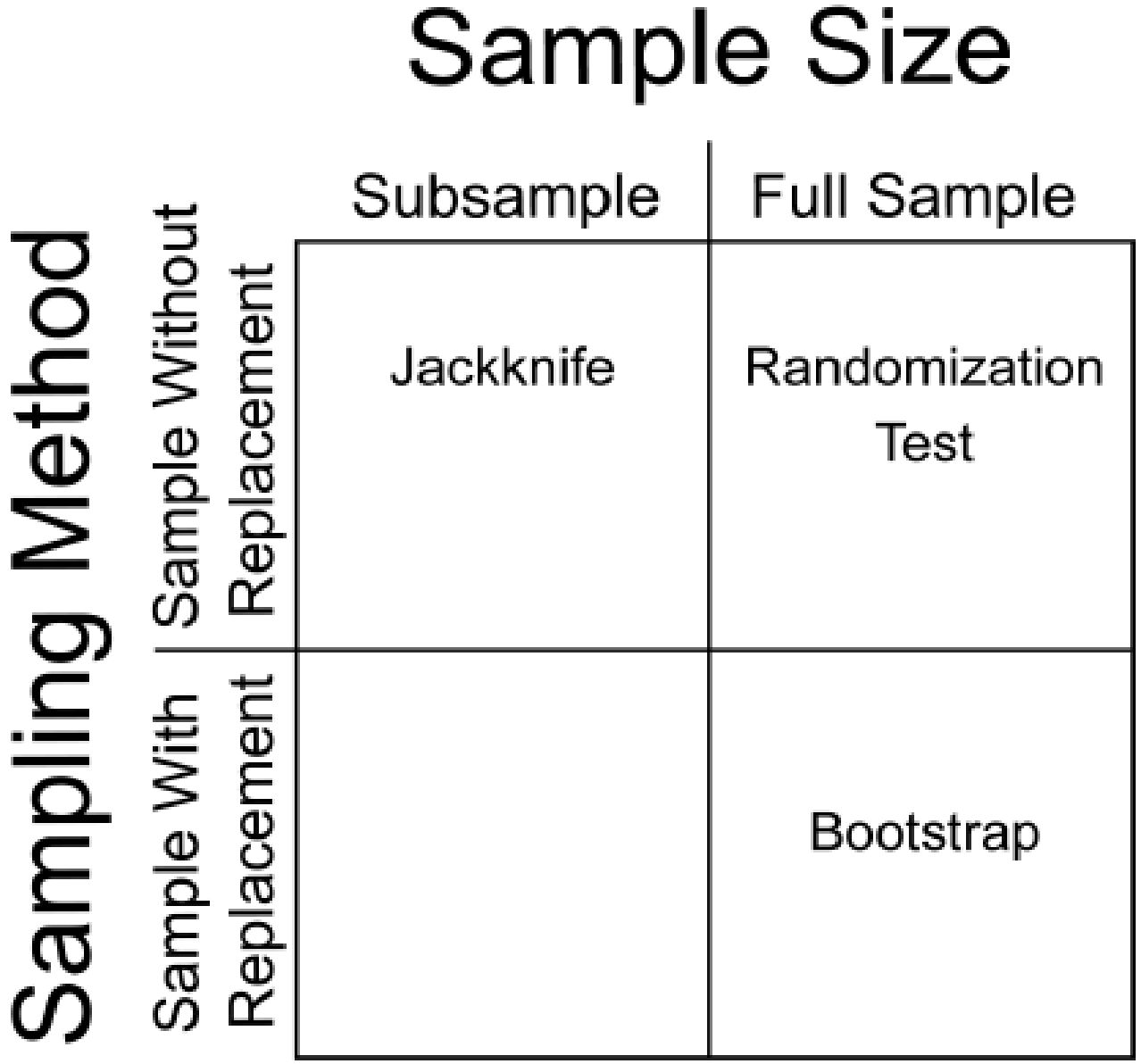
Failure of the Bootstrap

- If take $t = 0.0001$,

$$\lim_n P_{F_n}(T_n^* \leq t) \geq 1 - e^{-1}$$

$$\lim_n P_F(T_n \leq t) = 1 - e^{-0.0001} \approx 0$$

$$\text{So } P_{F_n}(T_n^* \leq t) \not\to P_F(T_n \leq t)$$


Sample Size
Subsample
Full Sample
Sampling Method
Sample Without Replacement
Sample With Replacement
Jackknife
Randomization Test
Bootstrap

Empirical likelihood

- Owen, A. (1990). Empirical Likelihood Ratio Confidence Regions. The Annals of Statistics, 18(1), 90-120.
- Empirical likelihood (EL) is a nonparametric approach for computing an estimator.
- Consider a random sample $X_1, \ldots, X_n$, of size n from some unknown distribution F.
- The natural choice for estimating the distribution function is the empirical distribution, that puts weight $1/n$ on each of the n sample points.

Empirical likelihood

- We do not specify any parametric model of *F*.
- We consider a PMF P such that

$$P(X_i) = F(X_i) - F(X_i^-) = W_i$$

- The likelihood function

$$L_n(P) = \prod_{i=1}^{n} P(X_i)$$

Constraint

- The log-likelihood function

$$\ell_n(P) = \sum_{i=1}^{n} \log P(X_i) = \sum_{i=1}^{n} \log W_i.$$

- The constraint

$$\mu_0 = \int x dF(x) = \sum_{i=1}^{n} P(X_i) X_i = \sum_{i=1}^{n} W_i X_i$$

Constrained optimization

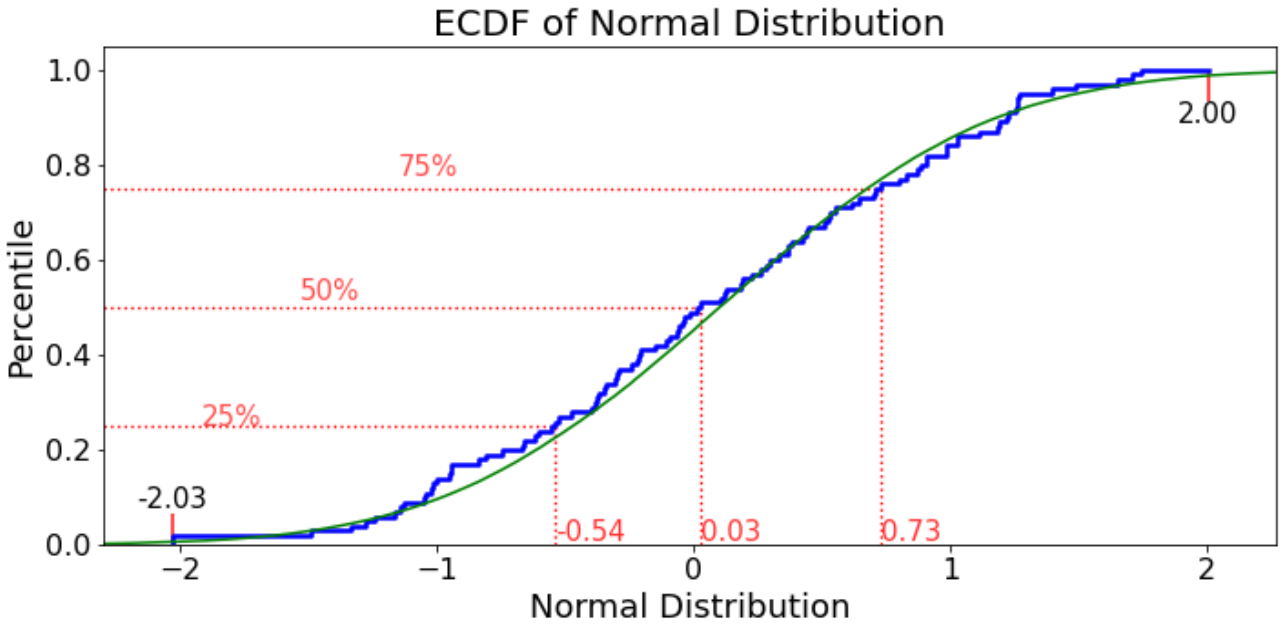
- MLE under the empirical likelihood

$$\widehat{W} = \text{argmax}_{W_1,\cdots,W_n} \sum_{i=1}^{n} \log W_i.$$

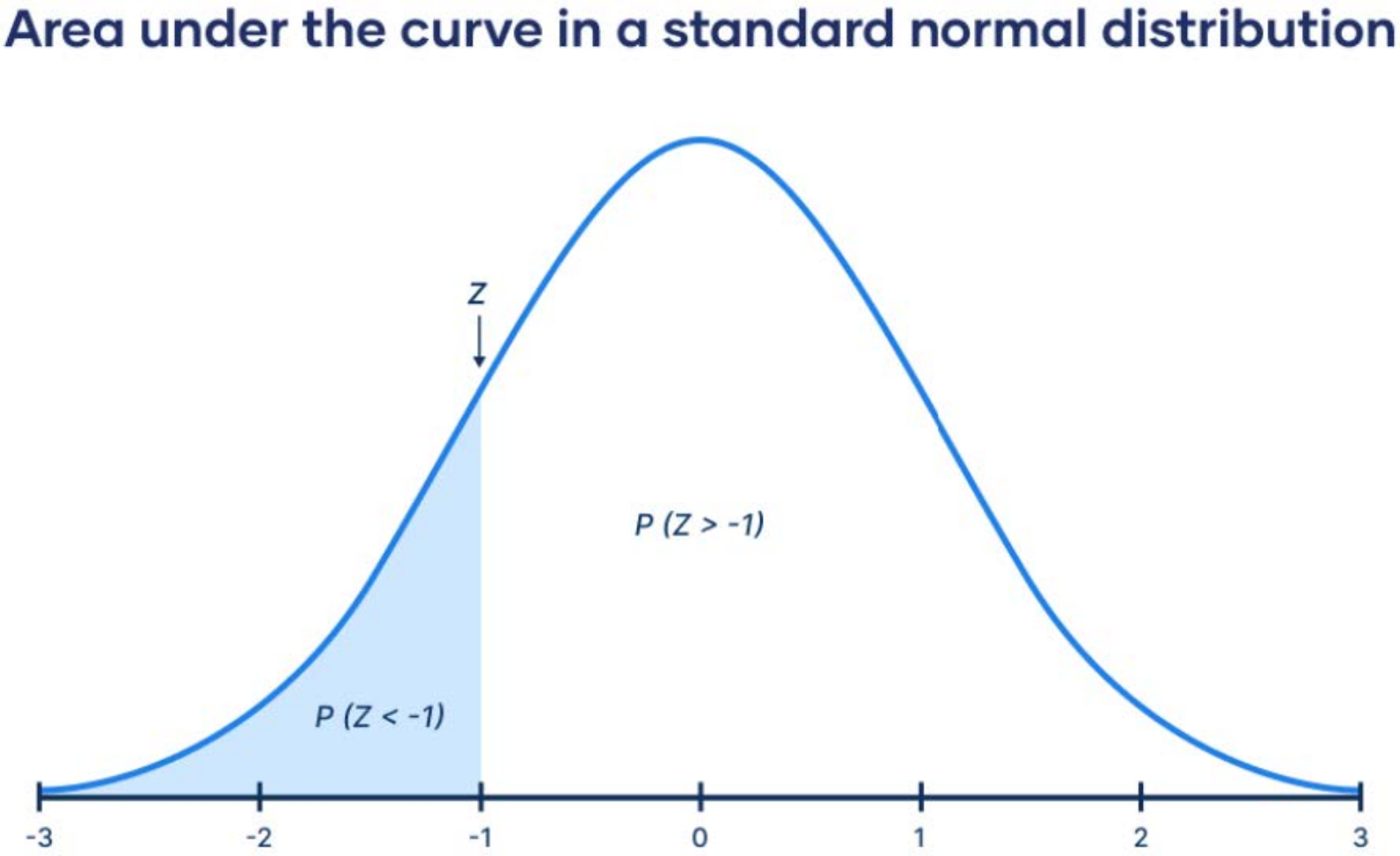
$$\text{s.t. } \mu_0 = \sum_{i=1}^{n} W_i X_i, \quad \sum_{i=1}^{n} W_i = 1, \quad W_i \geq 0.$$

MLE

- Using the Lagrangian multiplier method to maximize the logarithm of the empirical likelihood subject to the trivial normalization constraint, we find $W_i = 1/n$
- MLE of F is the empirical distribution function.



CDF of N(0, 1)


Area under the curve in a standard normal distribution
Z
P (Z > -1)
P (Z < -1)
-3
-2
-1
0
1
2
3

Lagrangian multiplier

- This procedure can be easily combined with any equation-type constraint

$$\widehat{W} = \text{argmax}_{W_1,\cdots,W_n} \sum_{i=1}^{n} \log W_i.$$

$$\text{s.t. } 0 = \sum_{i=1}^{n} W_i g(X_i), \quad \sum_{i=1}^{n} W_i = 1, \quad W_i \geq 0$$

- Using the Lagrangian multiplier

$$\sum_{i=1}^{n} \log W_i - \lambda^T \sum_{i=1}^{n} W_i g(X_i) - \mu(\sum_{i=1}^{n} W_i - 1)$$

Optimizer

- Taking derivative with respect to W_i

$$\frac{1}{W_i} = \lambda^T g(X_i) + \mu$$

$$W_i = \frac{1}{\mu + \lambda^T g(X_i)}.$$

$$n = \sum_{i=1}^{m} W_i(\mu + \lambda^T g(X_i)) = \mu + \lambda^T \underbrace{\sum_{i=1}^{n} W_i g(X_i)}_{=0} = \mu.$$

Solution of EL

- As a result, we obtain a closed-form solution as

$$W_i = \frac{1}{n + \widehat{\lambda}^T g(X_i)}$$

- The multiplier λ satisfies the constraint

$$0 = \sum_{i=1}^{n} \frac{g(X_i)}{n + \widehat{\lambda}^T g(X_i)}.$$

Bayesian Bootstrap

- Instead of drawing weights w_i from the discrete set $\{0, 1/n, \ldots, n/n\}$, the Bayesian approach treats the vector of weights w in F_n as unknown parameters and derives a posterior distribution for w, and hence $T(F)$.
- Rubin (1981) used a non-informative prior, $\prod_{i=1}^{n} w_i^{-1}$, which, when combined with the multinomial likelihood for Z, leads to a Dirichlet$(1, \ldots, 1)$ distribution for the posterior distribution of w.
- The posterior distribution of $T(F)$ is estimated by Monte Carlo method: generate $w^{(b)}$ from a Dirichlet$(1, \ldots, 1)$ distribution and then calculate $T(F_n^{(b)})$ for each sample $w^{(b)}$.

Dirichlet distribution

- A Dirichlet distribution is a distribution over the K-dimensional probability simplex:

$$\Delta_K = \{(\pi_1, \ldots, \pi_K) : \pi_k \geq 0, \sum_k \pi_k = 1\}$$

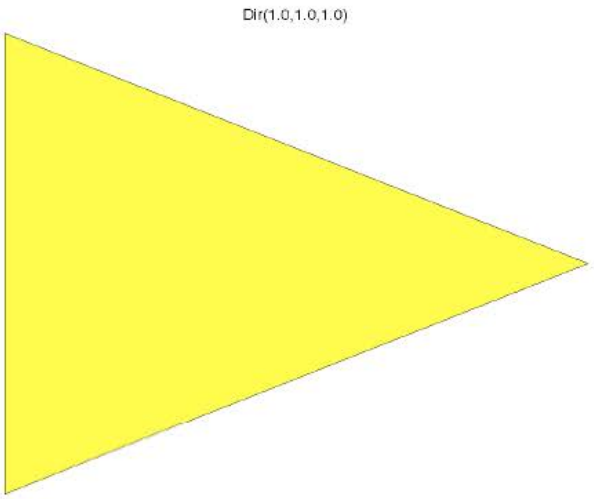
- We say $(\pi_1, \ldots, \pi_K)$ is Dirichlet distributed,

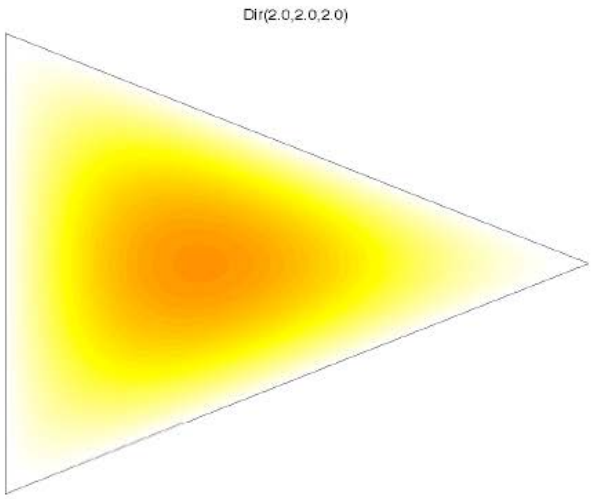
$$(\pi_1, \ldots, \pi_K) \sim \text{Dirichlet}(\alpha_1, \ldots, \alpha_K)$$

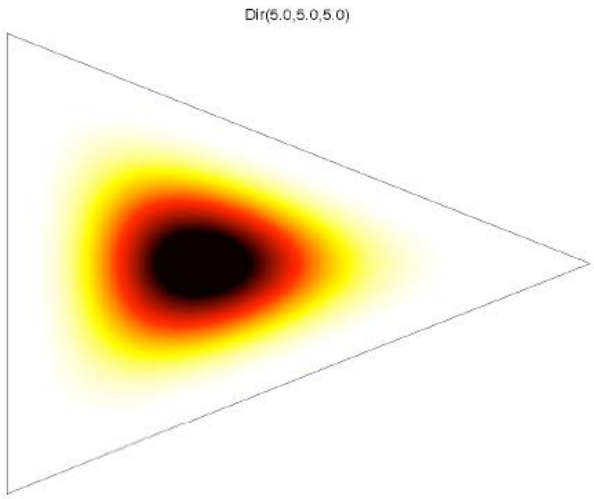
with parameters $(\alpha_1, \ldots, \alpha_K)$, if

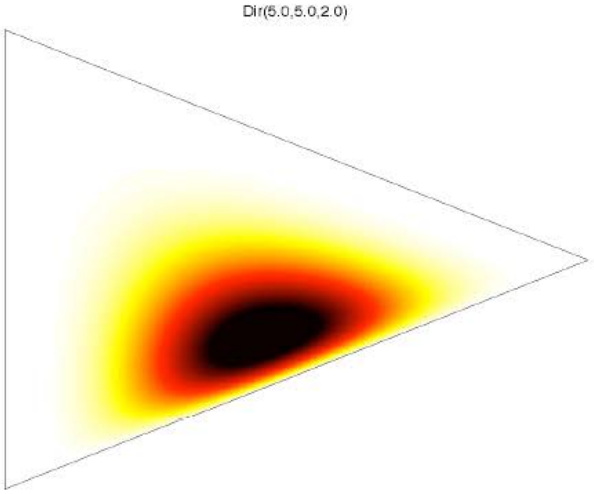
$$p(\pi_1, \ldots, \pi_K) = \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} \prod_{k=1}^{n} \pi_k^{\alpha_k - 1}$$

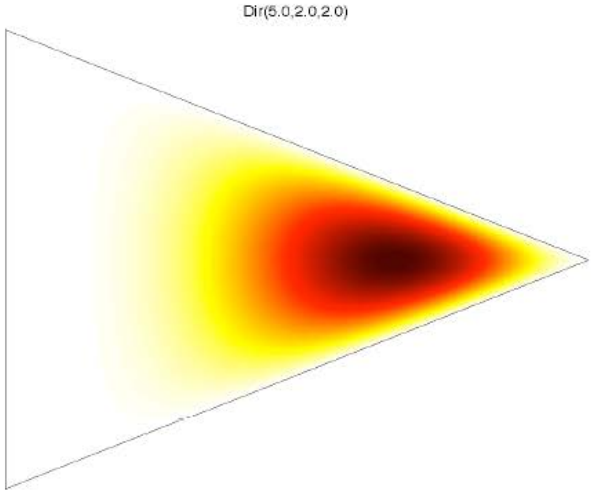
Examples of Dirichlet distributions

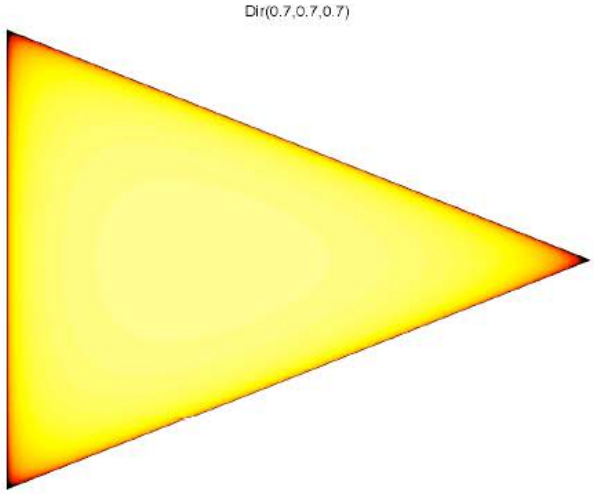












Likelihood and Prior

$$\Pr(x_i = d_k|\boldsymbol{\lambda}) = w_k, \quad \sum w_k = 1.$$

The $x_1, x_2, \ldots, x_n$ given w are assumed to be independent and identically distributed. Rubin (1981) showed that the Bayesian bootstrap procedure is equivalent to assuming that the prior distribution of w is the (improper) distribution

$$\Pr(\boldsymbol{\lambda}) = \prod_{k=1}^{K} w_k^{-1} \quad \text{if } \sum w_k = 1 \text{ and 0 otherwise.}$$

Posterior distribution of the weights

- The posterior distribution of the weights is

$$w \sim \text{Dirichlet}(1, \ldots, 1).$$

- The observations are fixed while the weights are varying (all observations are included with different weights),

$$\mu = \sum_{i=1}^{n} w_i \, y_i \, .$$

Two-step procedure

- Each Bayesian bootstrap sample is selected by a two-step procedure as follows:

Step 1. Draw n uniform random numbers between 0 and 1, and let their ordered values be $a_1, a_2, \ldots, a_n$; also let $a_0 = 0$ and $a_n = 1$.

Step 2. Draw each of the n values in $\mathbf{x}^{*b} = (x_1^{*b}, x_2^{*b}, \ldots, x_n^{*b})$ by drawing from $x_1, x_2, \ldots, x_n$ with probabilities $(a_1 - a_0), (a_2 - a_1), \ldots, (1 - a_{n-1})$; that is, independently n times, draw a uniform number u, and select x_i if $a_{i-1} < u \leq a_i$.

Block Bootstrap Methods

- If the underlying series is a stationary process with short-range dependence, then blocks of observations of suitable lengths should be approximately independent and the joint distribution of the variables in different blocks would be (about) the same, due to stationarity.
- If we resample blocks of observations, rather than observations one at a time, then that should bring us back to the nearly i.i.d. situation, a situation in which the bootstrap is known to succeed.

- Non-overlapping Block Bootstrap

$$B_1 = \{y_1, \ldots, y_h\},\ B_2 = \{y_{h+1}, \ldots, y_{2h}\}, \ldots, B_m = \{y_{(m-1)h+1}, \ldots, y_{mh}\}$$

- Moving Block Bootstrap

$$B_1 = \{y_1, \ldots, y_h\},\ B_2 = \{y_2, \ldots, y_{h+1}\}, \ldots, B_N = \{y_{n-h+1}, \ldots, y_n\}$$

- Circular Block Bootstrap: one periodically extends the observed series as

$$y_1, y_2, \ldots, y_n, y_1, y_2, \ldots, y_n, \ldots.$$

$$B_1 = \{z_1, \ldots, z_h\},\ B_2 = \{z_{h+1}, \ldots, z_{2h}\}, \ldots, B_n = \{z_n, \ldots, z_{n+h-1}\}.$$

Resampling Method by Perturbation

- A simple estimating function

$$S(\beta) = n^{-1} \sum_{i=1}^{n} Z_i \{Y_i - g(\beta' Z_i)\}$$

- Let G_i, $i = 1, \ldots, n$, be i.i.d. random variables with a known distribution whose mean and variance are one, Gamma(1, 1)=Exp(1),

$$S^*(\beta) = n^{-1} \sum_{i=1}^{n} z_i \{y_i - g(\beta' z_i)\} G_i = 0.$$

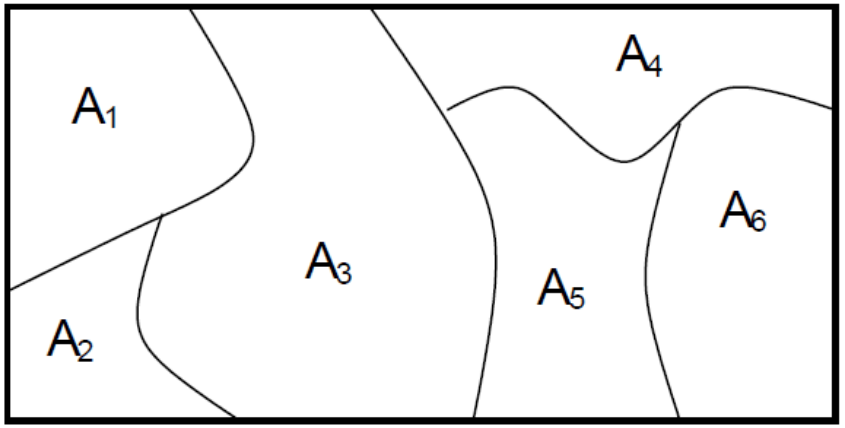
Dirichlet Process

A probability measure is a function from subsets of a space $\mathbb{X}$ to $[0, 1]$ satisfying certain properties.

A Dirichlet Process (DP) is a distribution over probability measures.

Denote $G \sim$ DP if G is a DP-distributed random probability measure.

For any finite set of partitions $A_1 \dot{\cup} \ldots \dot{\cup} A_K = \mathbb{X}$, we require $(G(A_1), \ldots, G(A_K))$ to be Dirichlet distributed.



Parameters of Dirichlet Process

A DP has two parameters:

- Base distribution H, which is like the *mean* of the DP.
- Strength parameter α, which is like an *inverse-variance* of the DP.

We write:

$$G \sim \text{DP}(\alpha, H)$$

if for any partition $(A_1, \ldots, A_K)$ of $\mathbb{X}$:

$$(G(A_1), \ldots, G(A_K)) \sim \text{Dirichlet}(\alpha H(A_1), \ldots, \alpha H(A_K))$$

The first two cumulants of the DP:

$$\text{Expectation:} \qquad \mathbb{E}[G(A)] = H(A)$$

$$\text{Variance:} \qquad \mathbb{V}[G(A)] = \frac{H(A)(1 - H(A))}{\alpha + 1}$$

where A is any measurable subset of $\mathbb{X}$.

A distribution over distributions

- A Dirichlet Process is also a distribution over distributions

Let G be Dirichlet Process distributed:

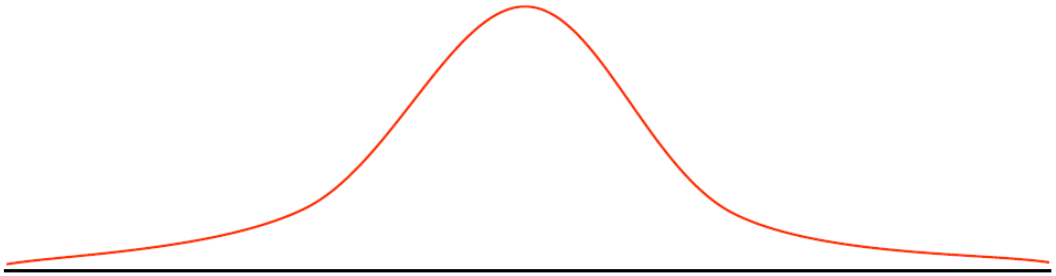
$$G \sim DP(\alpha, G_0)$$

- G_0 is a base distribution
- α is a positive scaling parameter

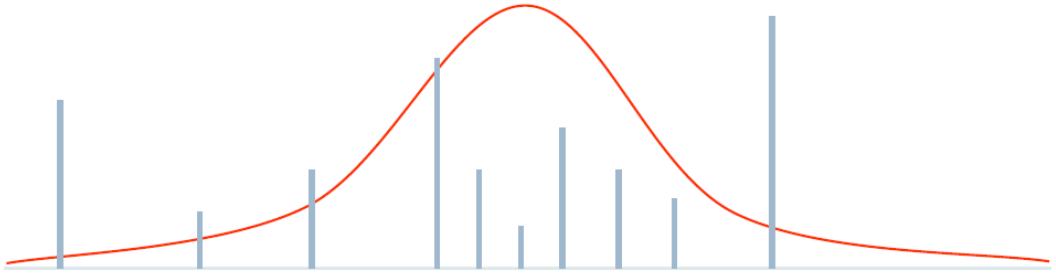
G is a random probability measure that has the same support as G_0

- G_0 is continuous, so the probability that any two samples are equal is precisely zero.
- G is a discrete distribution, made up of a countably infinite number of point masses, there is always a non-zero probability of two samples colliding

Consider Gaussian G_0



$G \sim DP(\alpha, G_0)$



Dirichlet Process

A partition of X is a collection of subsets $B_1, \ldots, B_N$ is such that, if $B_i \cap B_j = \emptyset$, $\forall i \neq j$ and $\cup_{i=1}^{N} B_i = X$.

Definition (Existence Theorem)

Let $\alpha > 0$ and H a probability distribution on X.
One can prove that there exists a unique distribution $DP(\alpha, H)$ on $\mathcal{F}(X)$ such that, if $F \sim DP(\alpha, H)$ and $B_1, \ldots, B_N$ is a partition of X then

$$(F(B_1), \ldots, F(B_N)) \sim \text{Dir}(\alpha H(B_1), \ldots, \alpha H(B_N)).$$

Mean and Variance of DP

Hereafter $F \sim DP(\alpha, H)$ and A is a measurable set in X.

- Expectation: $\mathbb{E}[F(A)] = \quad H(A)$.
- Variance: $\mathbb{V}[F(A)] = \frac{H(A)(1-H(A))}{\alpha+1}$

Posterior and Conjugacy

- Posterior and Conjugacy: let $x \sim F$ and consider a fixed partition $B_1, \ldots, B_N$, then

$$P(F(B_1), \ldots, F(B_N) | x \in B_k) =$$
$$\text{Dir}(\alpha H(B_1), \ldots, \alpha H(B_k) + 1, \ldots, \alpha H(B_N)).$$

It is possible to prove that if $S = (x_1, \ldots, x_n) \sim F$, and $F \sim DP(\alpha, H)$, then

$$P(F|S, \alpha, H) = DP\left(\alpha + n, \frac{1}{n+\alpha}\left(\alpha H + \sum_{i=1}^{n} \delta_{x_i}\right)\right)$$

Obtained distribution is discrete

From the form of the posterior we have that

$$\mathbb{E}(F(A)|S,\alpha,H) = \frac{1}{n+\alpha}\left(\alpha H(A) + \sum_{i=1}^{n}\delta_{x_i}(A)\right).$$

If $\alpha < \infty$ and $n \to \infty$ one can argue that

$$\mathbb{E}(F(A)|S,\alpha,H) = \sum_{i=1}^{\infty}\pi_i\delta_{x_i}(A)$$

where $(\pi_i)_{i=1}^{\infty}$ is the sequence corresponding to the limit $\lim_{n\to\infty} C_i/n$ of the empirical frequencies of the observations $(x_i)_{i=1}^{\infty}$.